Software Requirements Specification

(SRS)

Online Internship & Placement Preparation Portal

TEAM NO:- 6

Project	Online Internship & Placement Preparation Portal
Version	1.0
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Technology	MERN Stack (MongoDB, Express.js, React.js, Node.js) with Python for analytics
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Revision History

Version	Date	Author	Change Summary	Approval
1.0	6/9/2026	PES2UG24CS186,PES2UG24CS190, PES2UG24CS208, PES2UG24CS214	Initial SRS draft covering all functional and non-functional requirements	Pending

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Online Internship & Placement Preparation Portal (IPP), a web-based platform that connects students with internship opportunities, helps them prepare through aptitude/coding tests, and gives placement officers and recruiters tools to manage listings, applications, and placement analytics. It defines the functional and non-functional requirements, external interfaces, and verification criteria intended for developers, testers, and evaluators.

1.2 Scope

The system covers: student registration and profile management; browsing and searching internship listings; applying to internships with resume upload; taking timed aptitude/coding tests; tracking application and test progress through a personal dashboard; recruiter tools for posting listings and managing applications; and placement-officer/admin tools for approvals, user management, and institution-wide placement analytics. The system is built on the MERN stack (MongoDB, Express.js, React.js, Node.js), with a Python-based service handling test scoring and analytics computations. Out of scope: payroll/HR onboarding after selection, payment processing, and integration with third-party applicant tracking systems (ATS), which may be considered in a future release.

1.3 Audience

Developers, QA Engineers, Project Guide/Evaluators, Students, Recruiters, and Placement Cell Administrators.

1.4 Definitions, Acronyms and Abbreviations

- IPP — Internship & Placement Preparation Portal (this system)
- MERN — MongoDB, Express.js, React.js, Node.js
- JWT — JSON Web Token, used for authentication
- RBAC — Role-Based Access Control
- OTP — One-Time Password
- API — Application Programming Interface
- UI/UX — User Interface / User Experience
- CSV — Comma-Separated Values
- TLS — Transport Layer Security
- WCAG — Web Content Accessibility Guidelines

2. Overall Description

2.1 Product Perspective

IPP is a new, self-contained web application, not a modification of an existing system. It consists of a React.js single-page front end, a Node.js/Express REST API back end, a MongoDB document store for application data, and a Python microservice for aptitude-test scoring and analytics aggregation. It integrates with an email/SMS provider for notifications and OTP delivery, and with cloud/local file storage for resumes.

2.2 Major Product Functions (Detailed)

- Role-based registration and secure authentication (Student, Recruiter, Placement Officer/Admin)
- Internship listing creation, search, and filtering
- Resume upload and one-click application submission
- Application status tracking and interview scheduling
- Timed aptitude/coding test engine with auto-evaluation
- Personal progress dashboard for students
- Institution-wide placement analytics and reporting for admins
- Administrative approval workflow for recruiter accounts
- Notification system (in-app and email) for status changes

2.3 User Roles and Characteristics (Expanded)

- Student: primary end user; browses and applies to internships, takes aptitude tests, and tracks progress. Assumed to have basic web/mobile literacy.
- Recruiter/Company: posts internship listings and manages incoming applications; account requires admin approval before posting.
- Placement Officer / Admin: oversees the platform — approves recruiters, manages users, configures the test question bank, and reviews institution-wide analytics.

2.4 Operating Environment

Client: modern evergreen browsers (Chrome, Edge, Firefox, Safari) on desktop and mobile, responsive layout. Server: Node.js runtime on a cloud VM or container (e.g., AWS EC2/Render), MongoDB Atlas or self-hosted MongoDB, Python service deployed alongside the API for analytics. Requires stable internet connectivity for real-time features (notifications, live application status).

2.5 Constraints

- Must be built on the MERN stack with Python used only for the analytics/scoring microservice.
- Resume uploads limited to PDF, maximum 5MB.
- Must comply with institutional data-privacy norms for student academic and personal data.
- Aptitude test module must function correctly under intermittent network conditions (auto-save of progress).

3. External Interface Requirements

3.1 User Interfaces

A responsive React.js web UI with distinct dashboards for Student, Recruiter, and Admin roles. Accessibility features include keyboard navigation, screen-reader-friendly labels, and a high-contrast mode conforming to WCAG 2.1 AA where applicable.

3.2 Hardware Interfaces

- Client devices: desktop/laptop or mobile device with a modern web browser and internet access.
- Server infrastructure: cloud/on-premise servers hosting the Node.js API, MongoDB instance, and Python analytics service.
- No specialised hardware (card readers, dispensers, etc.) is required.

3.3 Software Interfaces

- Node.js/Express REST API exposing JSON endpoints over HTTPS to the React front end.
- MongoDB (via Mongoose ODM) for persistent storage of users, listings, applications, and test data.
- Python (Flask) analytics/scoring microservice, invoked via internal REST calls from the Node.js API.
- Third-party email/SMS API for OTP delivery and status-change notifications.
- Cloud object storage (or local disk in development) for resume and document files.

3.4 Communications

- All client-server and server-to-microservice traffic secured with TLS 1.2 or higher.
- REST API with JSON payloads; JWT bearer tokens carried in the Authorization header.
- Retry-with-backoff strategy for transient failures when calling the email/SMS provider or analytics microservice.

4. System Features (Detailed)

Each requirement below includes its acceptance criteria and a reference test case ID. Requirement IDs follow the pattern IPP-F-###.

4.1 Authentication & User Management

Description: Secure, role-based registration and login for students, recruiters, and placement officers.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
IPP-F-001	The system shall allow students, recruiters, and placement officers to register with role-based accounts using email verification.	Functional	High	Students / Recruiters / Admin	AC: Registration succeeds only after email OTP verification; account created with correct role. Test: TC-Auth-01	Requires email/OTP service
IPP-F-002	The system shall authenticate users via email/password and issue a signed JWT for session management.	Functional	High	Security	AC: Valid credentials return a JWT valid for the configured expiry; invalid credentials are rejected. Test: TC-Auth-02	JWT secret via env config
IPP-F-003	The system shall hash and salt all user passwords before storage and never store passwords in plaintext.	Functional	High	Security	AC: Database inspection shows only bcrypt/argon2 hashes. Test: TC-Auth-03	bcrypt/argon2 library
IPP-F-004	The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts and log the event.	Functional	Medium	Security	AC: 6th attempt is blocked with a lockout message; an audit-log entry is created. Test: TC-Auth-04	Rate limiter middleware
IPP-F-005	The system shall allow users to reset a forgotten password via a time-limited emailed reset link.	Functional	Medium	Students / Recruiters	AC: Reset link expires after 30 minutes; a reused link is rejected. Test: TC-Auth-05	Email service dependency

4.2 Internship Listings & Search

Description: Recruiters publish listings; students discover them via search and filters.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
IPP-F-006	The system shall allow verified recruiters to create, edit, and publish internship listings with role, stipend, duration, skills, and deadline.	Functional	High	Recruiters	AC: Listing is saved and becomes visible to students only after admin approval. Test: TC-List-01	Requires admin approval workflow
IPP-F-007	The system shall allow students to search and filter internship listings by domain, location, stipend, and duration.	Functional	High	Students	AC: Filtered results match all applied criteria and return within 5 seconds. Test: TC-List-02	MongoDB text/compound index
IPP-F-008	The system shall display detailed internship information including company profile, eligibility, and application deadline.	Functional	Medium	Students	AC: Detail page renders all required fields with no missing-data errors. Test: TC-List-03	
IPP-F-009	The system shall automatically hide a listing from active search results once its application deadline passes.	Functional	Medium	Recruiters / Admin	AC: Listing is hidden from active search results after the deadline timestamp. Test: TC-List-04	Scheduled job (cron)

4.3 Application Management

Description: Students apply to listings; recruiters review and update application status.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
IPP-F-010	The system shall allow students to upload a resume (PDF, max 5MB) and apply to an internship listing in one action.	Functional	High	Students	AC: An application record links student, resume, and listing; duplicate applications are blocked. Test: TC-App-01	File storage (cloud/local)
IPP-F-011	The system shall allow recruiters to view, filter, and update application status (Applied, Shortlisted, Rejected, Selected).	Functional	High	Recruiters	AC: Status change is persisted and reflected on the student dashboard in real time. Test: TC-App-02	
IPP-F-012	The system shall notify students by email/in-app notification whenever their application status changes.	Functional	High	Students	AC: Notification is generated within 5 minutes of the status change. Test: TC-App-03	Notification service
IPP-F-013	The system shall allow recruiters to schedule and record interview details (date, mode, link) for shortlisted candidates.	Functional	Medium	Recruiters	AC: Interview details are visible to the student on the application detail page. Test: TC-App-04	

4.4 Aptitude Test Module

Description: Timed, auto-evaluated aptitude/coding tests with a configurable question bank.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
IPP-F-014	The system shall provide timed aptitude/coding tests composed of MCQ and coding questions drawn from a configurable question bank.	Functional	High	Students / Admin	AC: Test auto-submits when the timer reaches zero; questions are randomized per attempt. Test: TC-Test-01	Question bank module
IPP-F-015	The system shall auto-evaluate multiple-choice responses and display the score immediately after submission.	Functional	High	Students	AC: Score matches the expected value for a known answer key in test data. Test: TC-Test-02	
IPP-F-016	The system shall prevent a student from reattempting a completed test unless explicitly re-enabled by an administrator.	Functional	Medium	Admin	AC: Reattempt is blocked with a message; admin override creates a new attempt record. Test: TC-Test-03	
IPP-F-017	The system shall record test analytics (accuracy, time per question, category-wise performance) for each attempt.	Functional	Medium	Students / Admin	AC: An analytics record is generated and queryable per student and per test. Test: TC-Test-04	Python analytics service

4.5 Progress Tracking & Analytics Dashboard

Description: Personal dashboards for students; aggregate placement analytics for admins.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
IPP-F-018	The system shall provide students with a dashboard summarizing applications submitted, interviews scheduled, and test scores.	Functional	High	Students	AC: Dashboard figures match underlying records for a seeded test account. Test: TC-Dash-01	
IPP-F-019	The system shall provide placement officers with aggregate analytics on placement rate, company-wise applications, and student performance trends.	Functional	High	Placement Officer	AC: Aggregate figures reconcile with raw application/test data. Test: TC-Dash-02	Python analytics service
IPP-F-020	The system shall generate visual charts (bar/line/pie) for key metrics covering at least the last 12 months of data.	Functional	Medium	Placement Officer	AC: Charts render correctly for date ranges with and without data gaps. Test: TC-Dash-03	Charting library
IPP-F-021	The system shall allow authorized users to export analytics data as CSV or PDF reports.	Functional	Medium	Placement Officer	AC: Exported file contains all filtered records with correct headers. Test: TC-Dash-04	

4.6 Admin & Placement Officer Module

Description: Platform governance — account approvals, RBAC, and audit logging.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
IPP-F-022	The system shall allow administrators to approve or reject recruiter/company registrations before they can post listings.	Functional	High	Admin	AC: An unapproved recruiter cannot create listings; approval unlocks posting rights. Test: TC-Admin-01	
IPP-F-023	The system shall provide role-based access control (RBAC) restricting each module to Student, Recruiter, or Admin roles.	Functional	High	Security / Admin	AC: An access attempt outside the assigned role returns HTTP 403 and is logged. Test: TC-Admin-02	
IPP-F-024	The system shall maintain an audit log of administrative actions (approvals, deletions, role changes) with timestamp and actor ID.	Functional	Medium	Admin	AC: An audit-log entry is created for every administrative action in the test suite. Test: TC-Admin-03	

5. Non-Functional Requirements

NFRs below are measurable and tied to test plans. Requirement IDs follow the pattern IPP-NF-###.

Req ID	Requirement	Category	Priority	Acceptance Criteria / Measurement
IPP-NF-001	Page load and API response time shall be ≤ 8 seconds for 90% of requests under normal load (up to 5 concurrent users).	Performance	High	90th percentile ≤ 8s in load test (k6/JMeter). Test: TC-Perf-01
IPP-NF-002	The system shall maintain 90% uptime monthly, excluding scheduled maintenance windows.	Reliability	High	Uptime monitoring report ≥ 90% per month. Test: Ops dashboard review
IPP-NF-003	All data in transit shall be encrypted via TLS 1.2+; sensitive fields (passwords, tokens) shall be encrypted/hashed at rest.	Security / Compliance	High	Security scan shows no plaintext sensitive data; SSL Labs grade A. Test: TC-Sec-01
IPP-NF-004	The web interface shall conform to WCAG 2.1 AA guidelines for color contrast and keyboard navigation.	Usability / Accessibility	Medium	Accessibility audit (Lighthouse/axe) passes with no critical violations. Test: TC-UX-01
IPP-NF-005	The API layer shall support horizontal scaling to handle at least a 2x increase in concurrent users without code changes.	Scalability	Medium	Load test at 2x baseline shows near-linear scaling with added instances. Test: TC-Scale-01
IPP-NF-006	The codebase shall maintain a modular MVC structure with ≥ 40% automated test coverage for core modules (auth, applications, tests).	Maintainability	Medium	CI coverage report ≥ 40% for core modules. Test: CI pipeline report

5.1 Security

5.1.1 Security Objectives

- Protect the confidentiality and integrity of student personal data, resumes, and academic records against unauthorized access or disclosure.
- Ensure only authenticated and appropriately authorized users (Student, Recruiter, Placement Officer/Admin) can perform the actions permitted to their role.
- Maintain a verifiable audit trail of sensitive operations — registration approvals, application status changes, and administrative actions — to support accountability and traceability.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
IPP-SR-001	TLS 1.2+ shall be mandatory for all client-server and server-to-microservice network connections.	Security	High	Network scan confirms no unencrypted endpoints. Test: TC-Sec-02
IPP-SR-002	The system shall issue JWT access tokens with a maximum lifetime of 60 minutes and support refresh tokens with revocation.	Security	High	Expired token is rejected; a revoked refresh token cannot mint a new access token. Test: TC-Sec-03
IPP-SR-003	The system shall validate and sanitize all user inputs on both client and server to prevent NoSQL injection and XSS attacks.	Security	High	Automated scan (OWASP ZAP) reports no high-severity injection vulnerabilities. Test: TC-Sec-04
IPP-SR-004	The system shall enforce rate limiting on authentication and file-upload endpoints to mitigate brute-force and denial-of-service attempts.	Security	Medium	Requests beyond the configured threshold return HTTP 429. Test: TC-Sec-05
IPP-SR-005	The system shall restrict uploaded resume files to PDF format and 5MB maximum, scanning for malicious content before storage.	Security	Medium	Non-PDF, oversized, or malicious file upload is rejected. Test: TC-Sec-06
IPP-SR-006	The system shall enforce role-based access control (RBAC) at the API layer so no endpoint is reachable by a role lacking explicit permission.	Security	High	Automated RBAC test suite confirms 100% of endpoints reject unauthorized roles. Test: TC-Sec-07

6. Quality Attributes & Acceptance Tests

Exit criteria for acceptance: all High-priority functional requirements are implemented and verified, no critical NFR failures remain open, and the RTM (Section 8) shows all mapped test cases passed.

Acceptance test suites: Authentication, Listings & Search, Application Management, Aptitude Test Engine, Analytics Dashboard, Admin/RBAC, Performance, Security, and Accessibility.

7. System Models and Diagrams

7.1 UML Use-Case Diagrams

Two use-case diagrams illustrate the primary actor interactions with the system: one for the Student actor, and one for the Recruiter/Company and Placement Officer (Admin) actors.

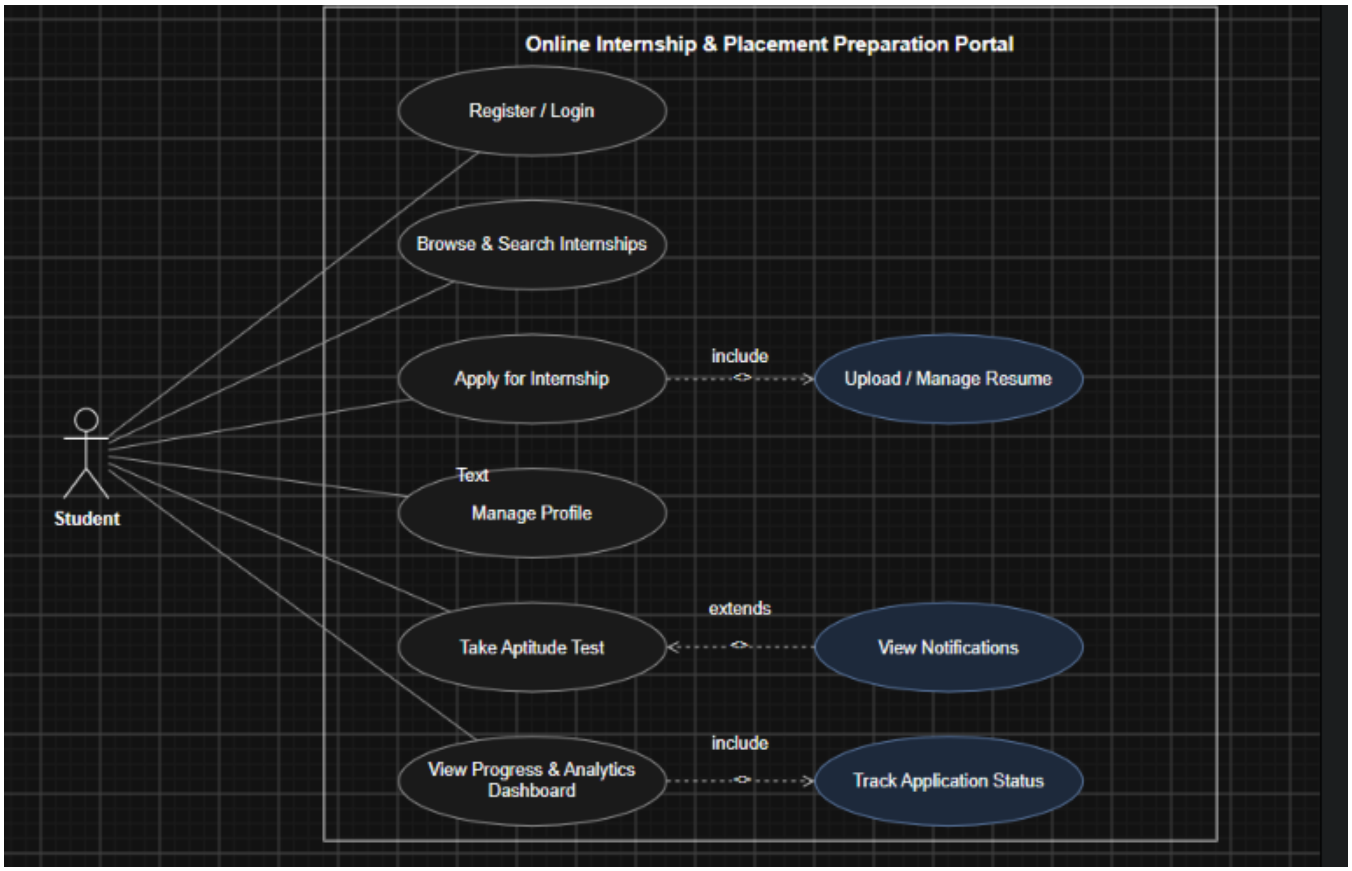


Figure 1: UML Use-Case Diagram — Student Interactions

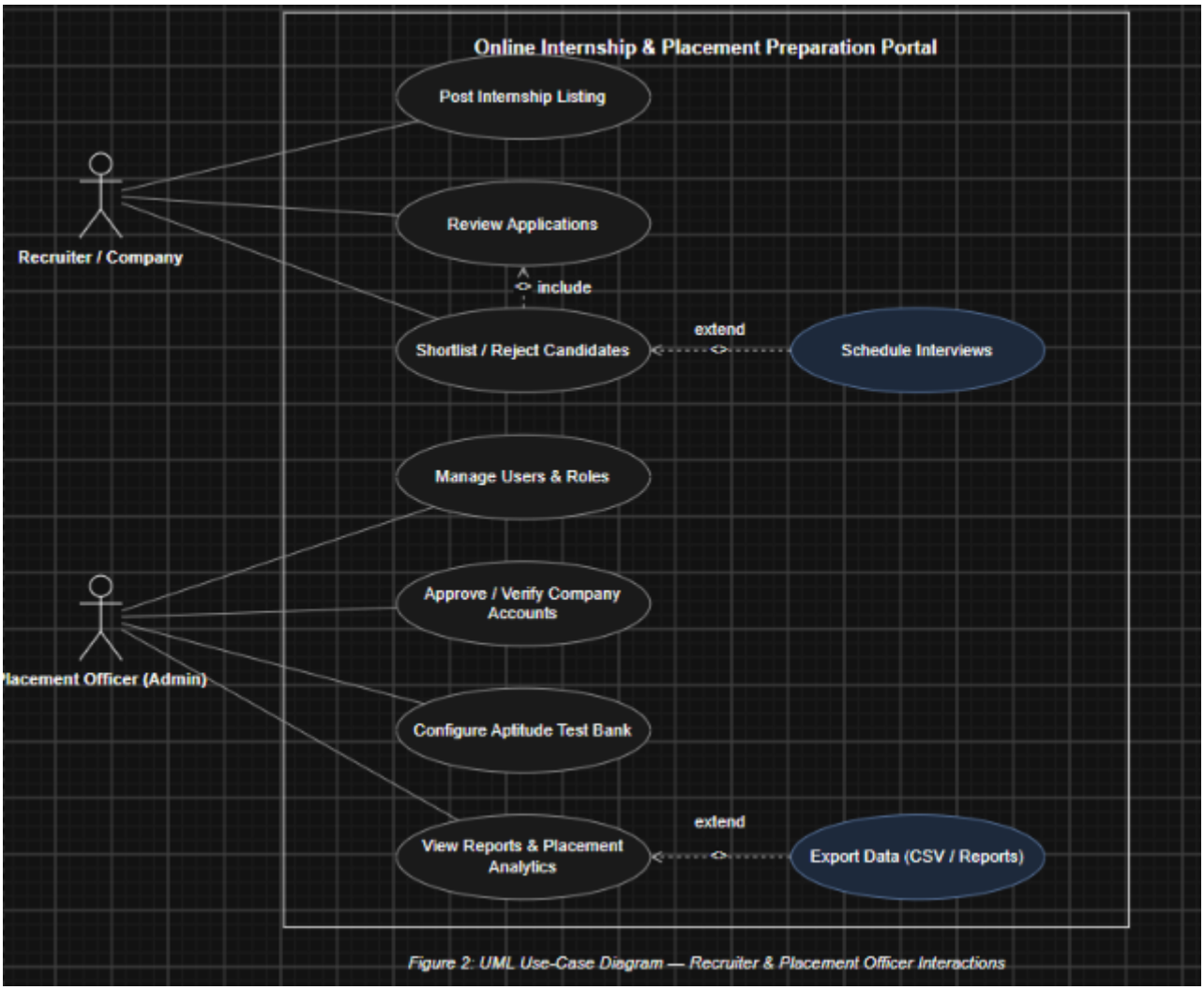


Figure 2: UML Use-Case Diagram — Recruiter & Placement Officer Interactions

8. Requirements Traceability Matrix (RTM)

Status legend: N = Not tested, P = In progress, A = Accepted/Passed.

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status(N/P/A)	Comments
IPP-F-001	Role-based registration	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
IPP-F-002	JWT authentication	4.1 / DS-Auth-01	AuthModule	TC-Auth-02	N	
IPP-F-006	Create internship listing	4.2 / DS-List-01	ListingModule	TC-List-01	N	
IPP-F-007	Search & filter listings	4.2 / DS-List-01	ListingModule	TC-List-02	N	
IPP-F-010	Apply to internship	4.3 / DS-App-01	ApplicationModule	TC-App-01	N	
IPP-F-011	Update application status	4.3 / DS-App-01	ApplicationModule	TC-App-02	N	
IPP-F-014	Timed aptitude test	4.4 / DS-Test-01	TestModule	TC-Test-01	N	
IPP-F-017	Test analytics recording	4.4 / DS-Test-01	AnalyticsModule	TC-Test-04	N	Python service

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status(N/P/A)	Comments
IPP-F-019	Placement analytics dashboard	4.5 / DS-Dash-01	AnalyticsModule	TC-Dash-02	N	Python service
IPP-F-022	Recruiter approval workflow	4.6 / DS-Admin-01	AdminModule	TC-Admin-01	N	
IPP-NF-001	API response time target	5 / DS-Perf-01	WebUI / API	TC-Perf-01	N	
IPP-NF-003	TLS & at-rest encryption	5 / DS-Sec-01	Security Layer	TC-Sec-01	N	
IPP-SR-002	JWT lifetime & revocation	5.1.2 / DS-Sec-02	AuthModule	TC-Sec-03	N	
IPP-SR-006	RBAC enforcement	5.1.2 / DS-Sec-03	Security Layer	TC-Sec-07	N	

RESPONSIBILITIES & CONTRIBUTIONS

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